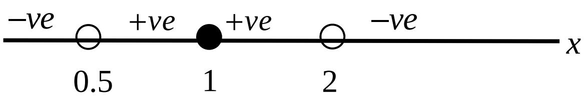
Yishun Junior College
2018 JC2 H2 Math Preliminary Examination
Paper 2 Solutions

Qns	Solutions
1	$\dfrac{x^2-2x+1}{-2+5x-2x^2} \leq 0$ $\dfrac{(x-1)^2}{(2x-1)(2-x)} \leq 0$  Therefore, $x<0.5$ or $x>2$ or $x=1$ Hence $\lvert x\rvert<0.5$ or $\lvert x\rvert>2$ or $\lvert x\rvert=1$ $-0.5<x<0.5$ or $x>2$ or $x<-2$ or $x=1$ or -1
2	$\mathrm{i}w+z=1$ $zw^*=2-6\mathrm{i}$ -- Eqn 2 $z=1-\mathrm{i}w$ -- Eqn 1 Subst. Eqn 1 into Eqn 2, $(1-\mathrm{i}w)w^*=2-6\mathrm{i}$ $w^*-\mathrm{i}ww^*=2-6\mathrm{i}$ Let $w=a+b\mathrm{i}$, $(a-b\mathrm{i})-\mathrm{i}(a+b\mathrm{i})(a-b\mathrm{i})=2-6\mathrm{i}$ $a-b\mathrm{i}-\mathrm{i}\left(a^2+b^2\right)=2-6\mathrm{i}$ $a+\mathrm{i}\left(-b-a^2-b^2\right)=2-6\mathrm{i}$ Comparing, $a=2$, $-b-2^2-b^2=-6$ $b^2+b-2=0$ $(b-1)(b+2)=0$ $b=1$ or $b=-2$ Therefore $w=2+\mathrm{i}$ or $2-2\mathrm{i}$ When $w=2+\mathrm{i}$, $z=1-\mathrm{i}(2+\mathrm{i})=2-2\mathrm{i}$ When $w=2-2\mathrm{i}$, $z=1-\mathrm{i}(2-2\mathrm{i})=-1-2\mathrm{i}$

3

(i) $\text{Rg} = (-1,1)$, $\text{D}_\text{f} = (-\infty,\infty) \setminus \{1,2\}$

$\text{Rg} \subseteq \text{D}_\text{f}$

$\therefore$ fg exists

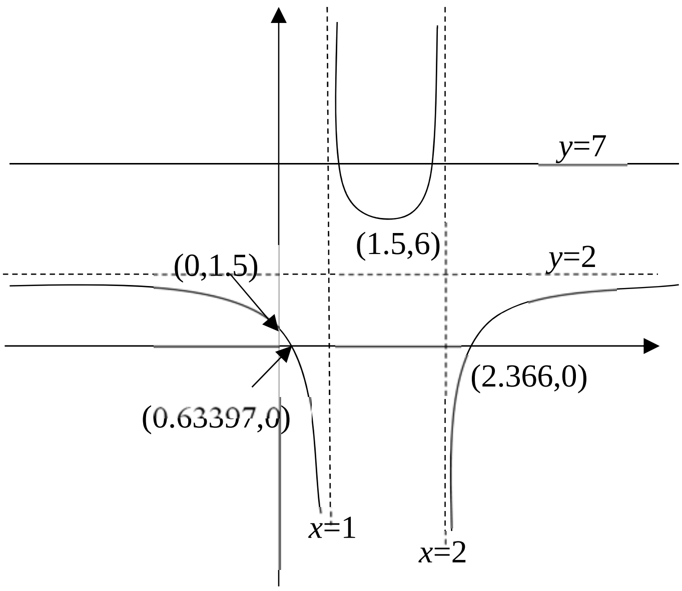
$$\text{fg}(x) = \text{f}(\sin x)$$

$$= 2 - \frac{1}{(\sin x - 1)(\sin x - 2)}$$

$$\text{D}_{\text{fg}} = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\text{R}_{\text{fg}} = \left(-\infty, \frac{11}{6}\right)$$

(ii) $\text{f}(x) = 2 - \dfrac{1}{(x-1)(x-2)}$



Since the horizontal line, $y = 7$ cuts the graph more than once, f is not one-one,
$\therefore$ f^{-1} does not exist.

(iii)

Largest value of a = $\dfrac{3}{2}$

(iv) Let $y = \mathrm{f}\left(x\right)$

$$y = 2 - \frac{1}{x^2 - 3x + 2}$$

$$\frac{1}{x^2 - 3x + 2} = 2 - y$$

$$x^2 - 3x + 2 = \frac{1}{2 - y}$$

$$x^2 - 3x + 2 - \frac{1}{2 - y} = 0$$

$$x = \frac{-\left(-3\right) \pm \sqrt{9 - 4\left(2 - \frac{1}{2 - y}\right)}}{2}$$

$$= \frac{3}{2} \pm \frac{1}{2}\sqrt{1 + \frac{4}{2 - y}}$$

Reject $x = \frac{3}{2} + \frac{1}{2}\sqrt{1 + \frac{4}{2 - y}}, \because D_{\mathrm{f}} = \left(-\infty, \frac{3}{2}\right)$

$\therefore \mathrm{f}^{-1} : x \mapsto \frac{3}{2} - \frac{1}{2}\sqrt{1 + \frac{4}{2 - x}}, x \in \mathbb{R}, x < 2$ or $x > 6$

4

(i)

$$\mathbf{a}\cdot\mathbf{n}=\mathbf{b}\cdot\mathbf{n}$$

$$\mathbf{a}\cdot\mathbf{n}-\mathbf{b}\cdot\mathbf{n}=0$$

$$(\mathbf{a}-\mathbf{b})\cdot\mathbf{n}=0$$

$$\overrightarrow{BA}\cdot\overrightarrow{ON}=0$$

Hence, *AB* is perpendicular to *ON*.

(ii)

$|\mathbf{a}\times\mathbf{n}|$ represents

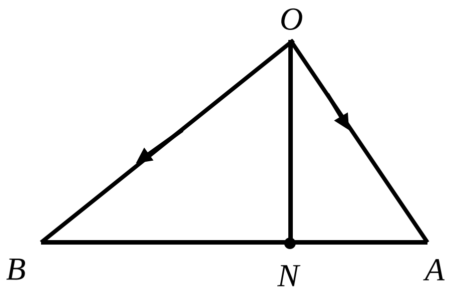
(1) twice area of triangle *OAN* (or)
(2) Area of parallelogram with adjacent sides *OA* and *ON*.

(iii)

$\left|\mathbf{a}\times\frac{\mathbf{n}}{|\mathbf{n}|}\right|$ represents

(1) the length of *AN* (or)

(2) (perpendicular) distance from *A* to *ON*



$$\tan 30^\circ=\frac{AN}{|\mathbf{n}|}$$

$$AN=\frac{1}{\sqrt{3}}|\mathbf{n}|$$

(iv)

$$\tan 60^\circ=\frac{NB}{|\mathbf{n}|}$$

$$NB=\sqrt{3}|\mathbf{n}|$$

$AN : NB$

$$=\frac{1}{\sqrt{3}}|\mathbf{n}| : \sqrt{3}|\mathbf{n}|$$

$$=1:3$$

By Ratio Theorem,

$$\mathbf{n}=\frac{3\mathbf{a}+\mathbf{b}}{4}$$

5	(a) $x=\mathrm{e}^t-1, y=t-\mathrm{e}^t$ $\frac{\mathrm{d}x}{\mathrm{d}t}=e^t\ ,\ \frac{\mathrm{dy}}{\mathrm{d}t}=1-e^t$ $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{1-e^t}{e^t}$ Eqn of normal: $y-\left(p-e^p\right)=-\ \frac{e^p}{1-e^p}(x-(e^p-1))$ When x = 0, $y-\left(p-e^p\right)=-\frac{e^p}{1-e^p}(0+1-e^p))$ $=-e^p$ $y=p-2e^p$ $Q(0,\ p-2e^p)$ $t-\mathrm{e}^t=-5$ From GC, $t=1.9368$ $\therefore\ \frac{\mathrm{dy}}{\mathrm{d}t}=1-e^{1.9368}$ $=\ -5.94$ unit / s (b) Let x = tan θ $\frac{\mathrm{d}x}{\mathrm{d}\theta}=\sec^2\theta$ When x = 0, tan $\theta = 0 \Rightarrow \theta = 0$ When $x=\sqrt{3}, \tan\theta=\sqrt{3}\Rightarrow\theta=\frac{\pi}{3}$ $\text{Vol}=\pi\int_0^{\sqrt{3}}\frac{x^2}{(1+x^2)^2}\ \mathrm{d}x=\pi\int_0^{\frac{\pi}{3}}\frac{\tan^2\theta}{(1+\tan^2\theta)^2}\sec^2\theta\ \mathrm{d}\theta$ $=\pi\int_0^{\frac{\pi}{3}}\frac{\tan^2\theta}{(\sec^2\theta)^2}\sec^2\theta\ \mathrm{d}\theta$ $=\pi\int_0^{\frac{\pi}{3}}\frac{\tan^2\theta}{\sec^2\theta}\ \mathrm{d}\theta$ $=\pi\int_0^{\frac{\pi}{3}}\frac{\sin^2\theta}{\cos^2\theta}\div\frac{1}{\cos^2\theta}\ \mathrm{d}\theta$

	$=\pi\int_0^{\frac{\pi}{3}}\sin^2\theta\ \mathrm{d}\theta$ $=\pi\int_0^{\frac{\pi}{3}}\frac{1-\cos 2\theta}{2}\ \mathrm{d}\theta$ $=\frac{\pi}{2}\left[\theta-\frac{\sin 2\theta}{2}\right]_0^{\frac{\pi}{3}}$ $=\frac{\pi}{2}\left[\frac{\pi}{3}-\frac{1}{2}\sin 2\left(\frac{\pi}{3}\right)-0\right]$ $=\frac{\pi^2}{6}-\frac{\sqrt{3}}{8}\pi$
6	(i) $\mathrm{P}(X=0)=\left(\frac{3}{5}\right)\left(\frac{2}{4}\right)(2)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)+\left(\frac{3}{5}\right)\left(\frac{2}{4}\right)\left(\frac{1}{2}\right)+\left(\frac{2}{5}\right)\left(\frac{1}{4}\right)\left(\frac{1}{2}\right)$ $=\frac{7}{20}$ (ii) $\mathrm{P}(X=2)$ $=\left(\frac{3}{5}\right)\left(\frac{2}{4}\right)(2)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)$ $=\frac{3}{20}$ $\mathrm{P}(X=1)$ $=1-\frac{3}{20}-\frac{7}{20}$ $=\frac{10}{20}$ Or $\mathrm{P}(X=1)$ $=\left(\frac{3}{5}\right)\left(\frac{2}{4}\right)(2)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)(2)+\left(\frac{3}{5}\right)\left(\frac{2}{4}\right)\left(\frac{1}{2}\right)+\left(\frac{2}{5}\right)\left(\frac{1}{4}\right)\left(\frac{1}{2}\right)$ $=\frac{10}{20}$ Probability distribution of *X* <table><tr><td>x</td><td>0</td><td>1</td><td>2</td></tr><tr><td>P(*X* = *x*)</td><td>$\frac{7}{20}$</td><td>$\frac{10}{20}$</td><td>$\frac{3}{20}$</td></tr></table> (iii) E(*X*) = $0\times\frac{7}{20}+1\times\frac{10}{20}+2\times\frac{3}{20}=\frac{16}{20}=\frac{4}{5}$

	Var(X) $= 0^2 \times \frac{7}{20} + 1^2 \times \frac{10}{20} + 2^2 \times \frac{3}{20} - \left(\frac{16}{20}\right)^2 = \frac{23}{50}$ (iv) P(2 balls are of different colour|no heads) $= \dfrac{\left(\frac{3}{5}\right)\left(\frac{2}{4}\right)(2)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)}{\frac{7}{20}}$ $= \frac{3}{7}$ (v) Required probability = $\frac{7}{20} + \left[\left(\frac{3}{5}\right)\left(\frac{2}{4}\right) + \left(\frac{2}{5}\right)\left(\frac{1}{4}\right)\right] - \left[\left(\frac{3}{5}\right)\left(\frac{2}{4}\right)\left(\frac{1}{2}\right) + \left(\frac{2}{5}\right)\left(\frac{1}{4}\right)\left(\frac{1}{2}\right)\right]$ $= \frac{11}{20}$
7	(i) No of ways $= \frac{{}^{10}C_5}{2} = 126$ (ii) No of ways $= 4! \times 5! \times 3! = 17280$ (iii) No of ways $= {}^{10}C_5 \times 5!$ $= 30240$ (iv) No of ways no two odd digit cards are next to each other $= (7-1)! \times {}^7P_3$ $= 151200$ Required probability = $\frac{151200}{362880} = \frac{5}{12}$
8	(i) Large number of glowing sticks with constant probability of 0.12 enables independence in selection. (ii) Let X be the number of green glowing sticks out of 8. $X \sim \mathrm{B}(8, 0.12)$ $\mathrm{P}(X \geq 3) = 1 - \mathrm{P}(X \leq 2)$ $= 0.060789...$ ≈ 0.0608 (3 s.f) (iii) Let Y be the number of packets that contain at least 3 glowing sticks out of 40 packets. $Y \sim \mathrm{B}(40, 0.060789)$ $\mathrm{P}(Y < 6) = \mathrm{P}(Y \leq 5)$ $= 0.96733...$ ≈ 0.967 (3 s.f)

(iv) Let W be the number of blue glowing sticks out of 8.

$W \sim \mathrm{B}(8\,,\ p)$

Since 3 is the modal number,

$\mathrm{P}(W=3) > \mathrm{P}(W=2)$

$\binom{8}{3} p^3(1-p)^5 > \binom{8}{4} p^2(1-p)^6$ --- (1)

$56p^3(1-p)^5 > 28p^2(1-p)^6$

$2p > 1-p$

$p > \frac{1}{3}$

At the same time,

$\mathrm{P}(W=3) > \mathrm{P}(W=4)$

$\binom{8}{3} p^3(1-p)^5 > \binom{8}{4} p^4(1-p)^4$ --- (2)

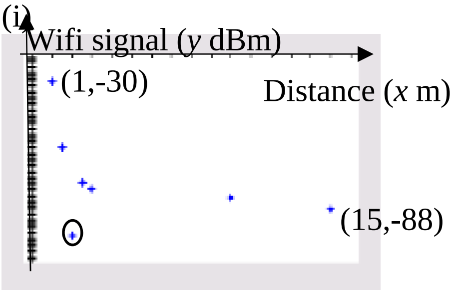
$56p^3(1-p)^5 > 70p^4(1-p)^4$

$\frac{4}{5}(1-p) > p$

$p < \frac{4}{9}$

Thus, $\frac{1}{3} < p < \frac{4}{9}$

9 (i)



There could be an obstruction which reduces the strength of the wifi signal or
There could be a wifi disruption.

(ii) The data points do not seem to lie on a straight line.

(iii) For $y = \frac{c}{x} + d$, $r = 0.97034... \approx 0.970$

For $y = \frac{e}{x^2} + f$, $r = 0.99733... \approx 0.997$

	Since the r value for the second model is closer to 1, $y=\dfrac{e}{x^2}+f$ is the better model. $e=55.81195...\approx 55.8\ (3\text{ s.f})$ $f=-85.42578...\approx -85.4\ (3\text{ s.f})$ (iv) When $x=5,\ y=\dfrac{55.81195}{5^2}-85.42578$ $=-83.193...\approx -83.2\ (3\text{ s.f})$ The required signal strength is -83.2 dBm This estimate is expected to be reliable as it is obtained by interpolation and $r\approx 0.997$ is close to 1.
10	(i) 'Keep the recorded values small since they are around 49' or 'give an indication of the variations around the hypothesized mean of 49. (ii) Unbiased estimate of μ, $\bar{x}=\dfrac{\sum(x-49)}{100}+49$ $=\dfrac{313}{100}+49$ $=52.13$ Unbiased estimate of σ^2, s^2 $=\dfrac{1}{100-1}\left(\sum(x-49)^2-\dfrac{\left[\sum(x-49)\right]^2}{100}\right)$ $=\dfrac{1}{99}\left(23280-\dfrac{(313)^2}{100}\right)$ $=225.26$ ≈ 225 (iii) $H_0:\mu=49$ $H_1:\mu\neq 49$ Under H_0, the test statistic is $Z=\dfrac{\overline{X}-\mu}{s/\sqrt{n}}\sim \mathrm{N}(0,1)$ approximately (by CLT) where μ = 49, $\bar{x}=52.13$, n = 100, $s^2=225.26$ p- value = 0.037027 ≈ 0.0370 Since the p-value < 0.05 (the significance level), we reject H_0 and conclude that at the 5% level, there is significant evidence that the Exam Board's claim is not valid.

(iv) No assumptions needed since n = 100 is large, by Central Limit Theorem, the distribution of the sample mean marks is approximately normal.

(v) $H_0 : \mu = 49$

$H_1 : \mu > 49$

Under H_0, the test statistic is $Z = \dfrac{\overline{X} - \mu}{\sigma / \sqrt{n}} \sim \mathrm{N}(0,1)$

where μ = 59.2, n = 50, $\sigma = 11.0$

Critical region: $z > 1.7507$

Since the mean mark of candidates has not increased, we do not reject $H_0 \Rightarrow$

$$\frac{k-\mu}{\sigma / \sqrt{n}} < 1.7507$$

$$\frac{k-49}{11.0 / \sqrt{50}} < 1.7507$$

$$k < 51.723$$

$$\{k \in \mathrm{R} : k \le 51.7\}$$

11

(i) Let X be the lifespan of a light bulb manufactured by OceanDoor

$$X \sim \mathrm{N}\left(800, 70^2\right)$$

Let Y be the lifespan of a light bulb manufactured by DaRice

$$Y \sim \mathrm{N}\left(700, 80^2\right)$$

$$\overline{Y} = \frac{Y_1 + Y_2 + \cdots + Y_6}{6}$$

$$\overline{Y} \sim \mathrm{N}\left(700, \frac{80^2}{6}\right)$$

$$\mathrm{P}\left(\left|\overline{Y} - \frac{1}{2}X\right| \ge 200\right)$$

$$= 1 - \mathrm{P}\left(\left|\overline{Y} - \frac{1}{2}X\right| < 200\right)$$

$$= 1 - \mathrm{P}\left(-200 < \overline{Y} - \frac{1}{2}X < 200\right)$$

$$\overline{Y} - \frac{1}{2}X \sim \mathrm{N}\left(300, \frac{6875}{3}\right)$$

$$\mathrm{P}\left(\left|\overline{Y} - \frac{1}{2}X\right| \ge 200\right) = 0.982 \text{ (3sf)}$$

(ii)

$\mathrm{P}\left(\overline{X} \leq 825\right) \geq 0.99$

$\overline{X} \sim \mathrm{N}\left(800, \frac{70^2}{n}\right)$

$\mathrm{P}\left(Z \leq \frac{825-800}{70/\sqrt{n}}\right) \geq 0.99$

$\mathrm{P}\left(Z \leq \frac{5\sqrt{n}}{14}\right) \geq 0.99$

$\therefore \frac{5\sqrt{n}}{14} \geq 2.3263$

$n \geq 42.4275$

Or:

When $n = 42$, $\mathrm{P}\left(\overline{X} \leq 825\right) = 0.9897$

When $n = 43$, $\mathrm{P}\left(\overline{X} \leq 825\right) = 0.9904$

Least $n = 43$

(iii)

$X + Y \sim \mathrm{N}(1500, 11300)$

$\mathrm{P}(X + Y > 1450) = 0.681$ (3 sf)

(iv)

$\mathrm{P}(X > 725 \text{ and } Y > 725)$

$= \mathrm{P}(X > 725) \times \mathrm{P}(Y > 725)$

$= 0.323753$

$= 0.324$ (3 sf)

(v)

Part (iii) include cases that part (iv) does not have.

(vi) The lifespan of all light bulbs are independent of each other.